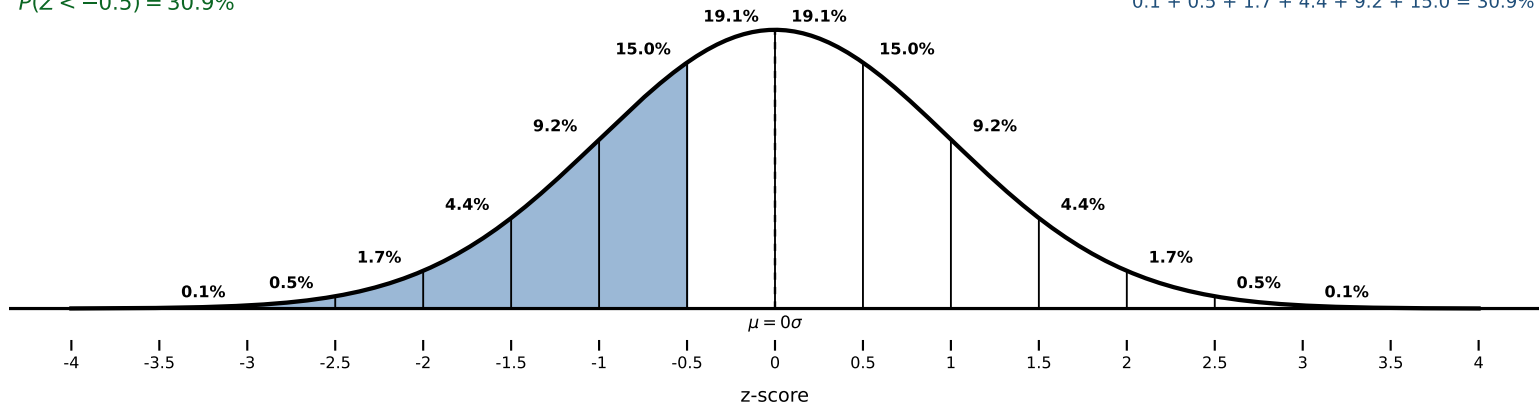


Question 5 — Shade & state the proportion of the distribution (68-95-99.7 rule refined to the nearest $\frac{1}{2}\sigma$)

(a) “Left-tailed” — $z = -0.5$ ($X = 3.675$ bushels)

$$P(Z < -0.5) = 30.9\%$$

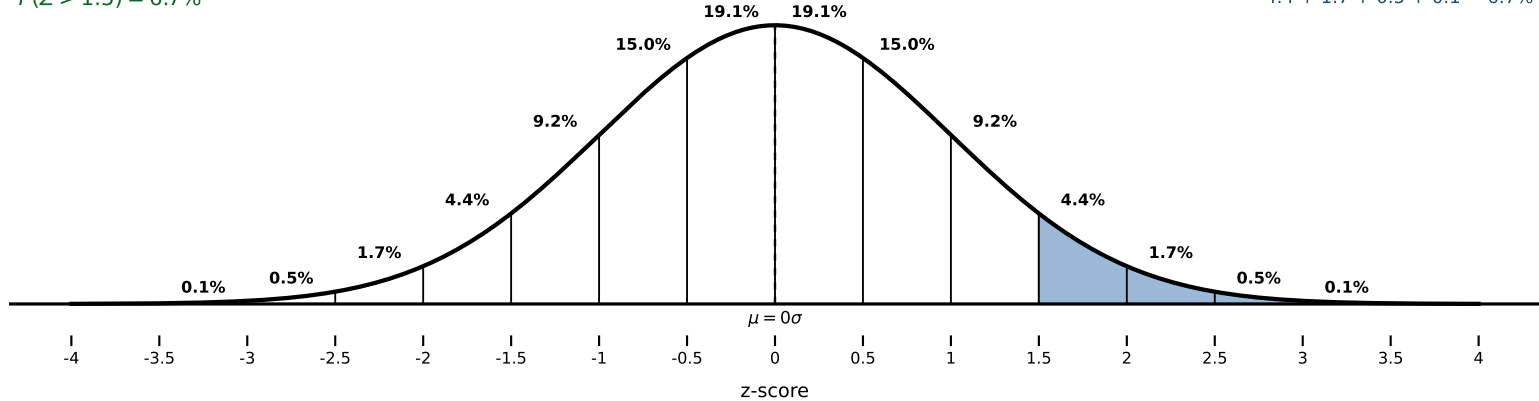
$$0.1 + 0.5 + 1.7 + 4.4 + 9.2 + 15.0 = 30.9\%$$



(b) “Right-tailed” — $z = 1.5$ ($X = 4.575$ bushels)

$$P(Z > 1.5) = 6.7\%$$

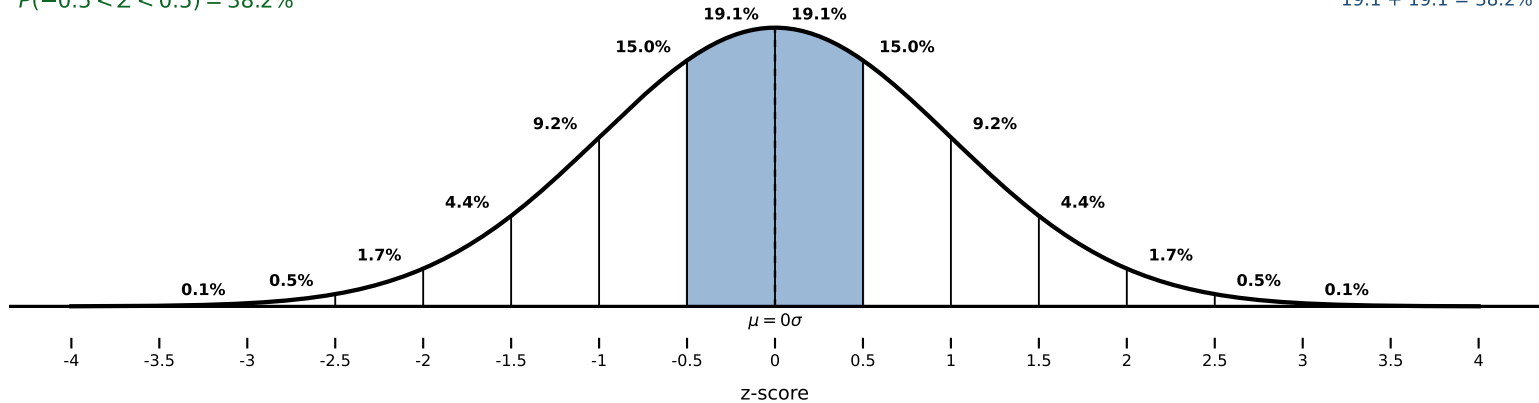
$$4.4 + 1.7 + 0.5 + 0.1 = 6.7\%$$



(c) “Centred” — $z = \pm 0.5$ ($X = 3.675$ & 4.125 bushels)

$$P(-0.5 < Z < 0.5) = 38.2\%$$

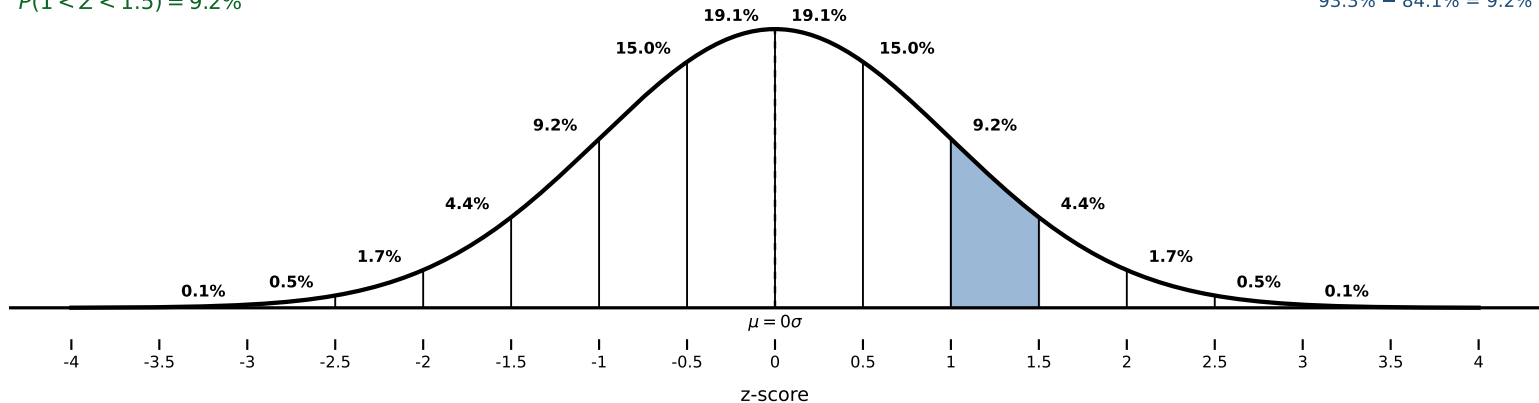
$$19.1 + 19.1 = 38.2\%$$



(d) “Bounded” — $1.0 < z < 1.5$ ($X = 4.35$ & 4.575 bushels)

$$P(1 < Z < 1.5) = 9.2\%$$

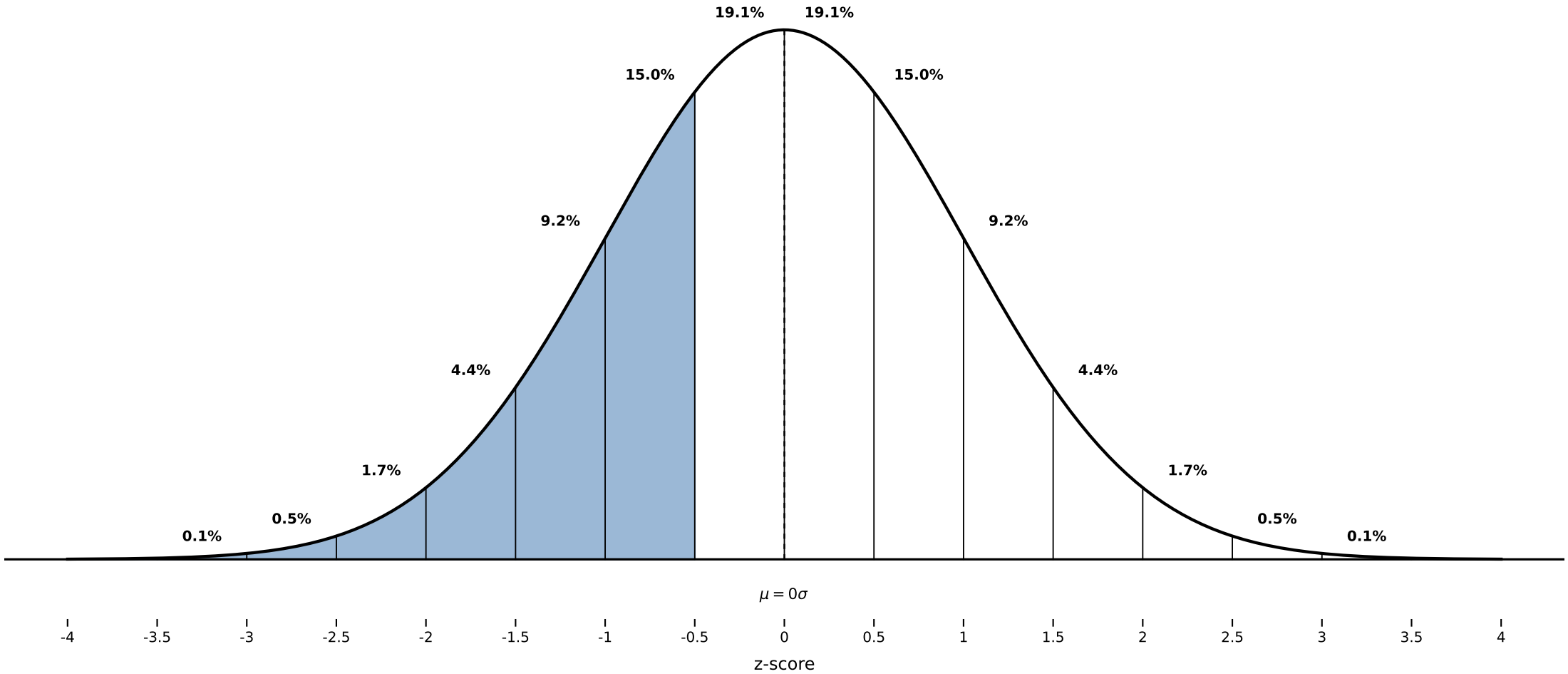
$$93.3 - 84.1 = 9.2\%$$



(a) “Left-tailed” — $z = -0.5$ ($X = 3.675$ bushels)

$P(Z < -0.5) = 30.9\%$

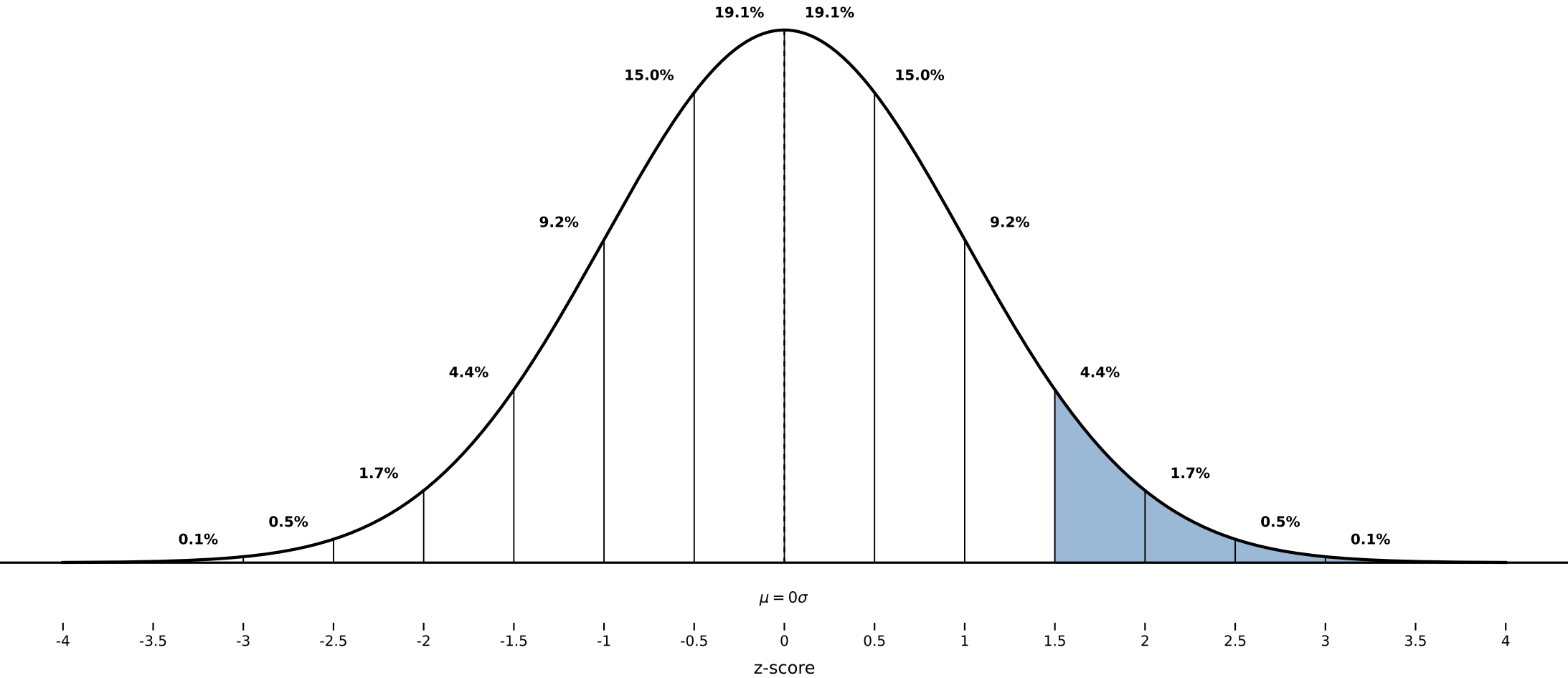
$0.1 + 0.5 + 1.7 + 4.4 + 9.2 + 15.0 = 30.9\%$



(b) “Right-tailed” — z = 1.5 (X = 4.575 bushels)

$P(Z > 1.5) = 6.7\%$

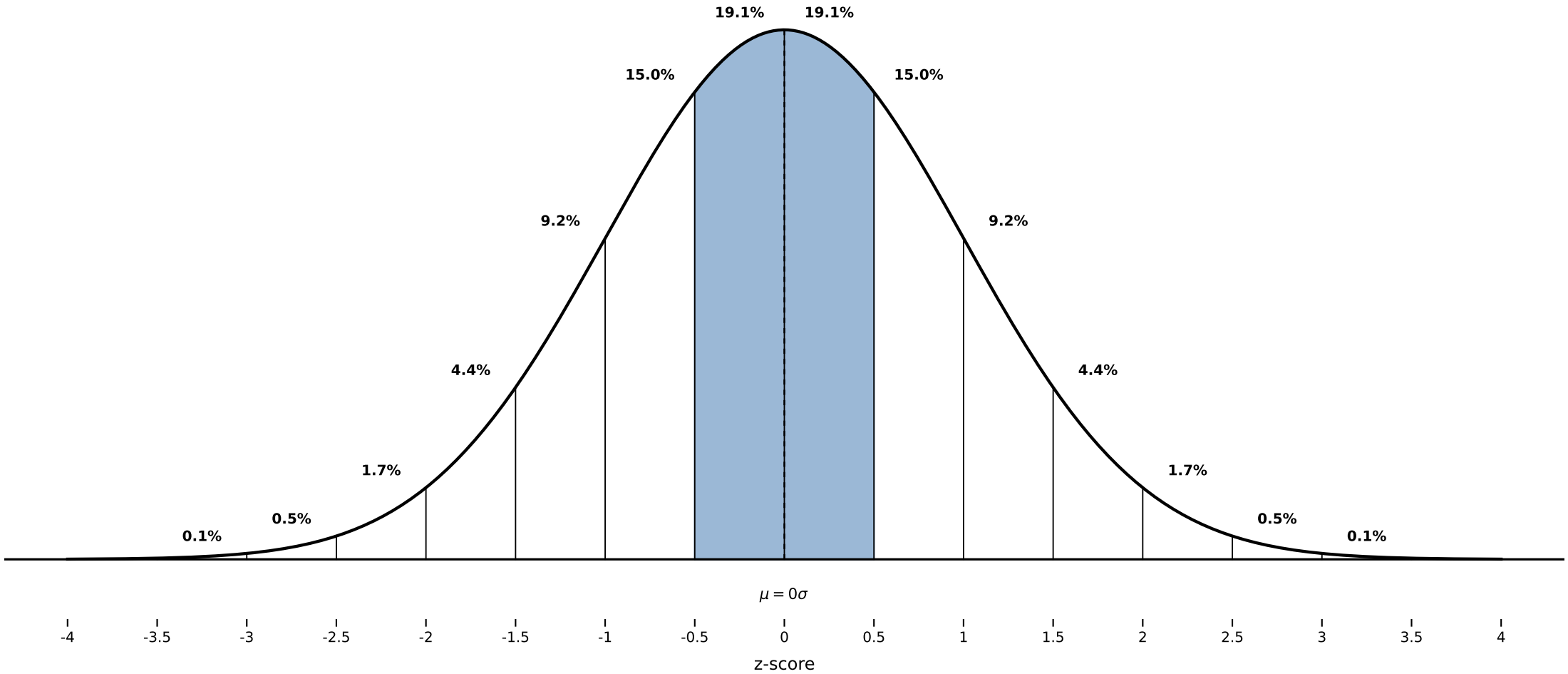
$4.4 + 1.7 + 0.5 + 0.1 = 6.7\%$



(c) “Centred” — $z = \pm 0.5$ ($X = 3.675$ & 4.125 bushels)

$P(-0.5 < Z < 0.5) = 38.2\%$

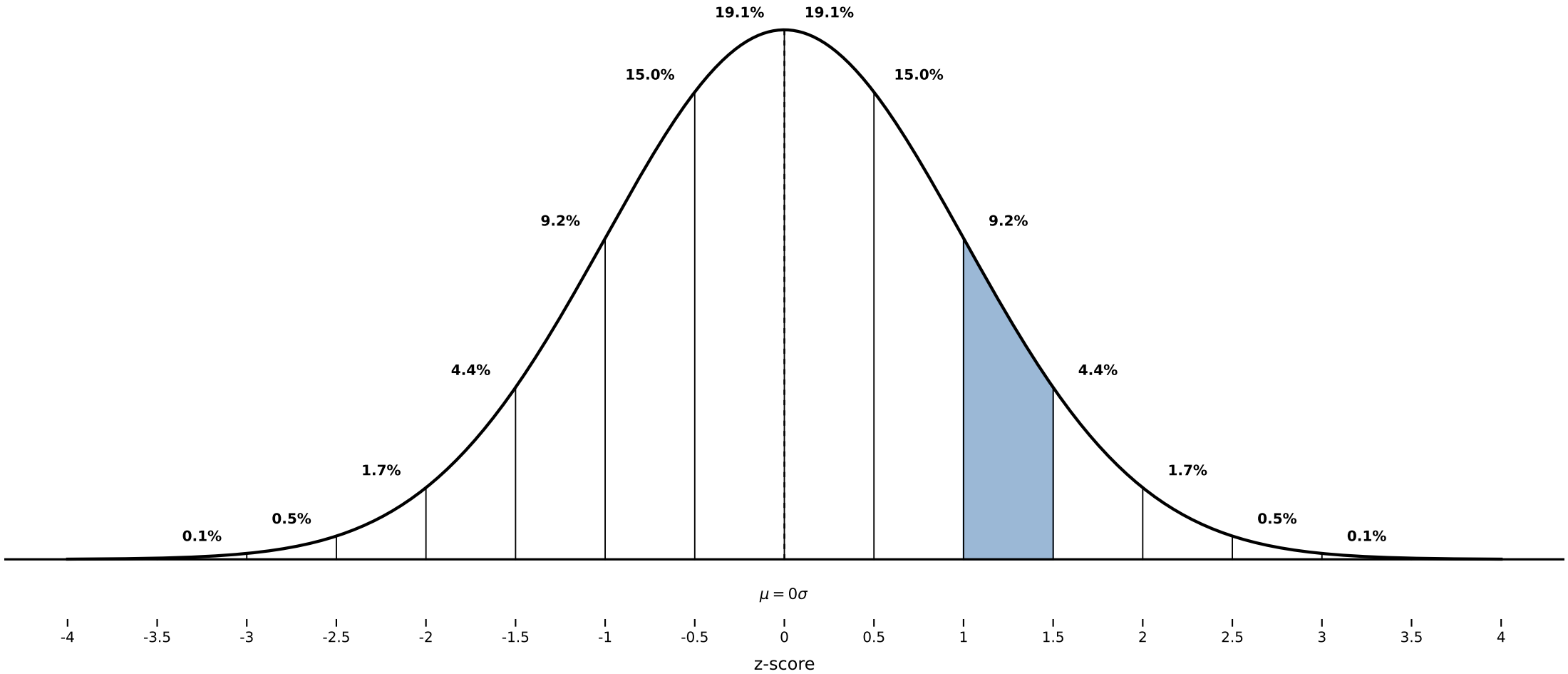
$19.1 + 19.1 = 38.2\%$



(d) “Bounded” — $1.0 < z < 1.5$ ($X = 4.35$ & 4.575 bushels)

$P(1 < Z < 1.5) = 9.2\%$

$93.3\% - 84.1\% = 9.2\%$



Questions 4 & 5 — Worked answers

Question 4

(a) **Standardising formula:**

$$z = \frac{X - \mu}{\sigma}$$

(b) **Using $\mu = 3.9$ bushels and $\sigma = 0.45$ bushels:**

- (i) $X = 4.125 \Rightarrow z = (4.125 - 3.9)/0.45 = 0.225/0.45 = +0.5$
- (ii) $X = 4.35 \Rightarrow z = (4.35 - 3.9)/0.45 = 0.45/0.45 = +1.0$
- (iii) $X = 3.675 \Rightarrow z = (3.675 - 3.9)/0.45 = -0.225/0.45 = -0.5$
- (iv) $X = 4.575 \Rightarrow z = (4.575 - 3.9)/0.45 = 0.675/0.45 = +1.5$

Question 5

(a) **Left-tailed** — $z = -0.5$ ($X = 3.675$ bushels)

Bands from the left tail up to -0.5 :

$$0.1 + 0.5 + 1.7 + 4.4 + 9.2 + 15.0$$

$$P(Z < -0.5) = 30.9\%$$

(b) **Right-tailed** — $z = 1.5$ ($X = 4.575$ bushels)

Bands from 1.5 out to the right tail:

$$4.4 + 1.7 + 0.5 + 0.1$$

$$P(Z > 1.5) = 6.7\%$$

(c) **Centred** — $z = \pm 0.5$ ($X = 3.675$ & 4.125 bushels)

The two central bands either side of μ :

$$19.1 + 19.1$$

$$P(-0.5 < Z < 0.5) = 38.2\%$$

(d) **Bounded** — $1.0 < z < 1.5$ ($X = 4.35$ & 4.575 bushels)

The single band between the two marks:

$$93.3\% - 84.1\%$$

$$P(1 < Z < 1.5) = 9.2\%$$

Cumulative reference table (area to the left of each $\frac{1}{2}\sigma$ mark)

-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5
≈0%	0.1%	0.6%	2.3%	6.7%	15.9%	30.9%	50%	69.1%	84.1%	93.3%	97.7%	99.4%	99.9%	≈100%

Top row: z Bottom row: area to the left, as a percentage

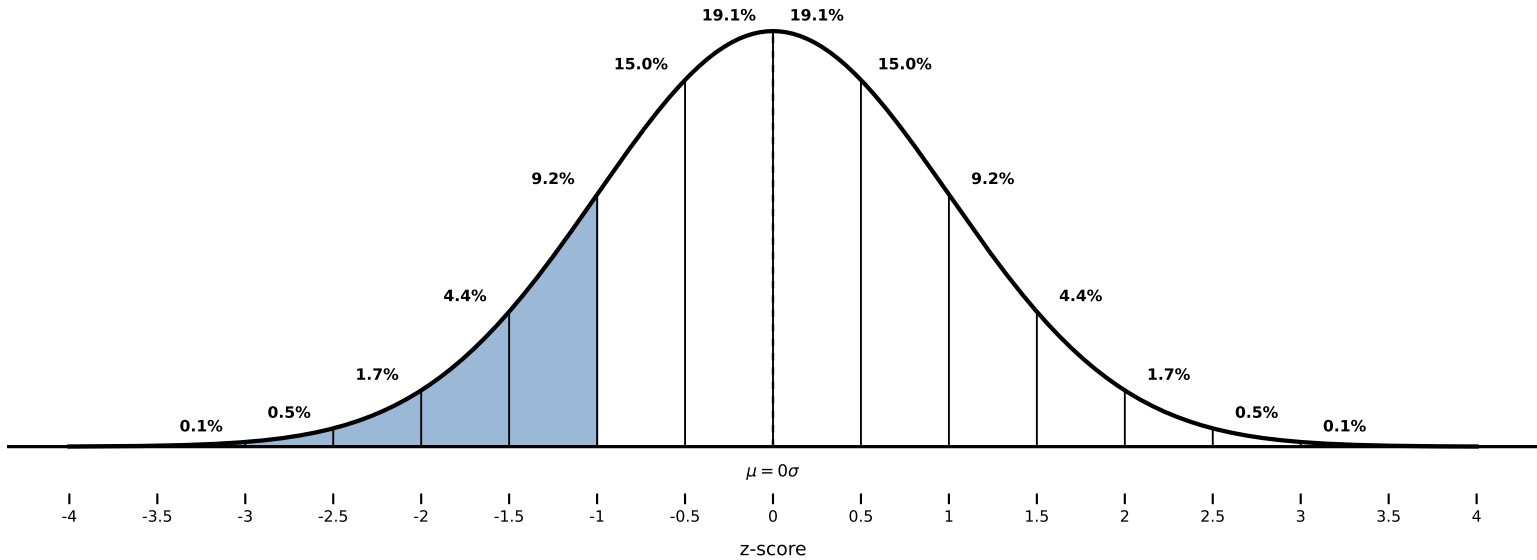
Note: because each band is rounded to 1 d.p., totals may differ from calculator values by about 0.1%.

Question 5(e) — Two crop yields with $z = -1$ and $z = 1.5$

(e)(i) a harvest yield of z-score < -1 ($X = 3.45$ bushels)

$$P(Z < -1) = 15.9\%$$

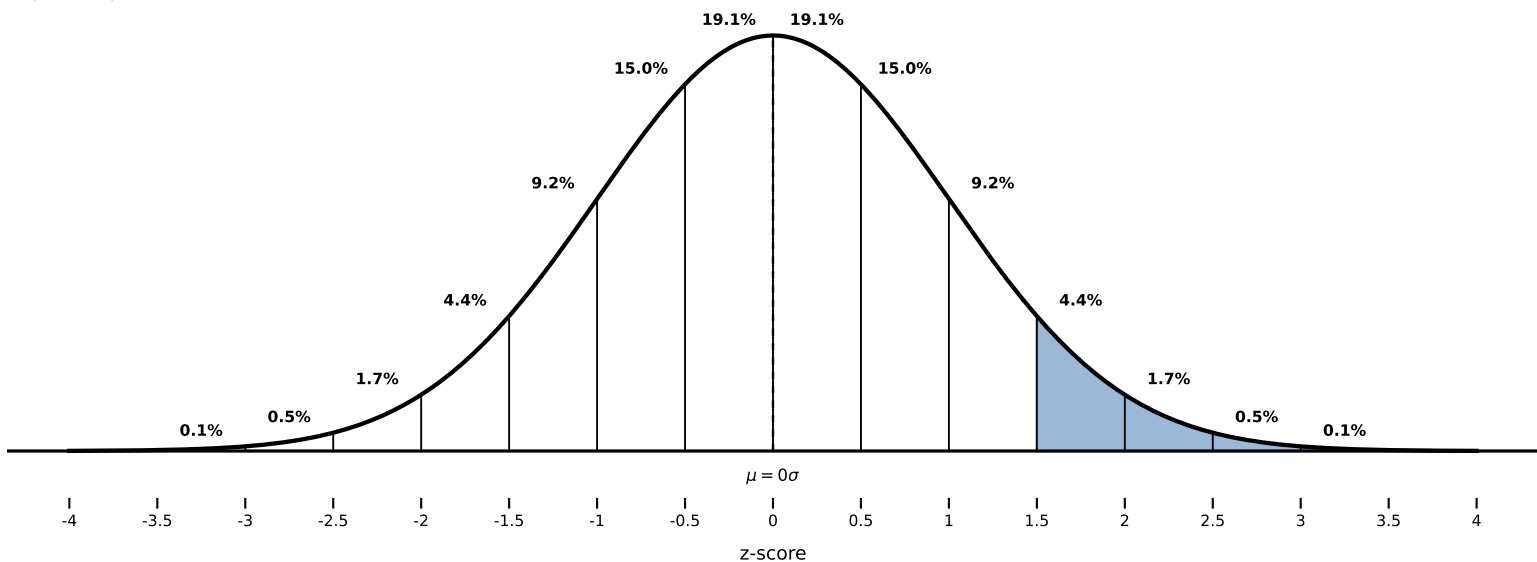
$$0.1 + 0.5 + 1.7 + 4.4 + 9.2 = 15.9\%$$



(e)(ii) a harvest yield of z-score > 1.5 ($X = 4.575$ bushels)

$$P(Z > 1.5) = 6.7\%$$

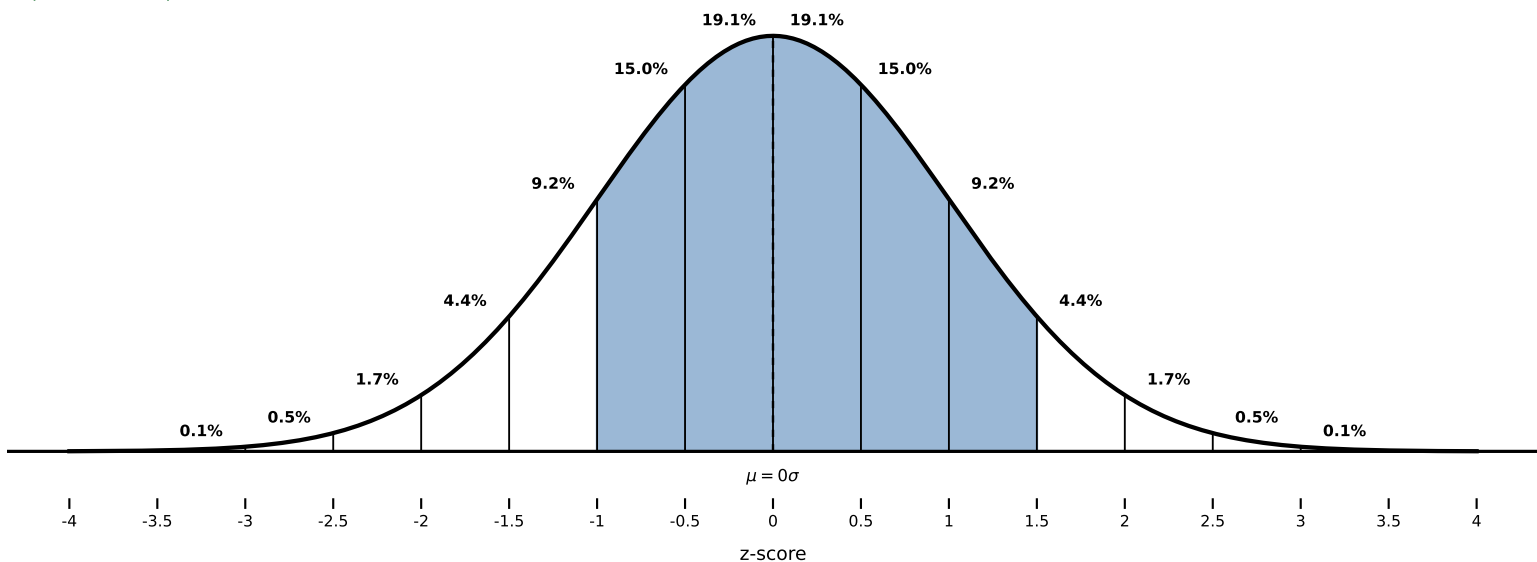
$$4.4 + 1.7 + 0.5 + 0.1 = 6.7\%$$



(e)(iii) a harvest yield with $-1 < \text{z-score} < 1.5$ (X between 3.45 and 4.575 bushels)

$$P(-1 < Z < 1.5) = 77.4\%$$

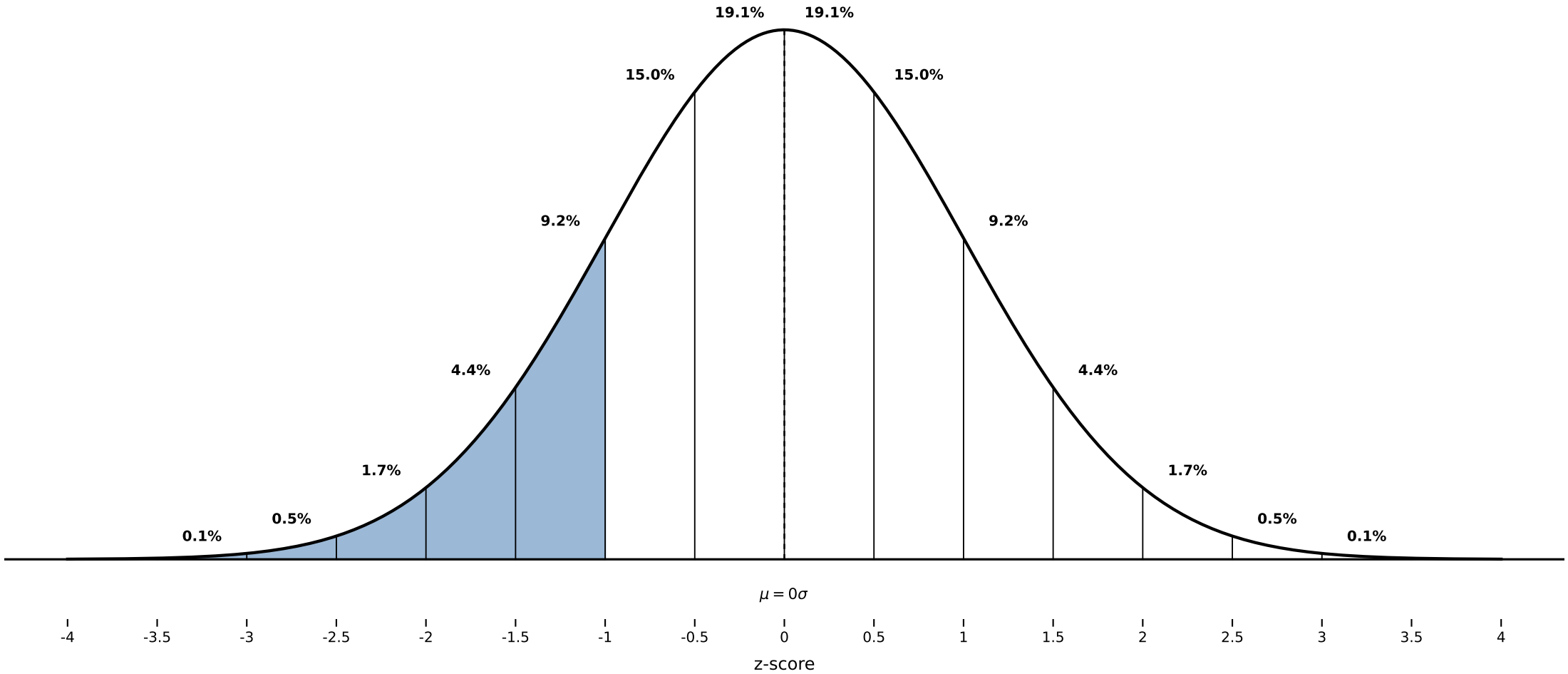
$$93.3\% - 15.9\% = 77.4\%$$



(e)(i) a harvest yield of z-score < -1 (X = 3.45 bushels)

$P(Z < -1) = 15.9\%$

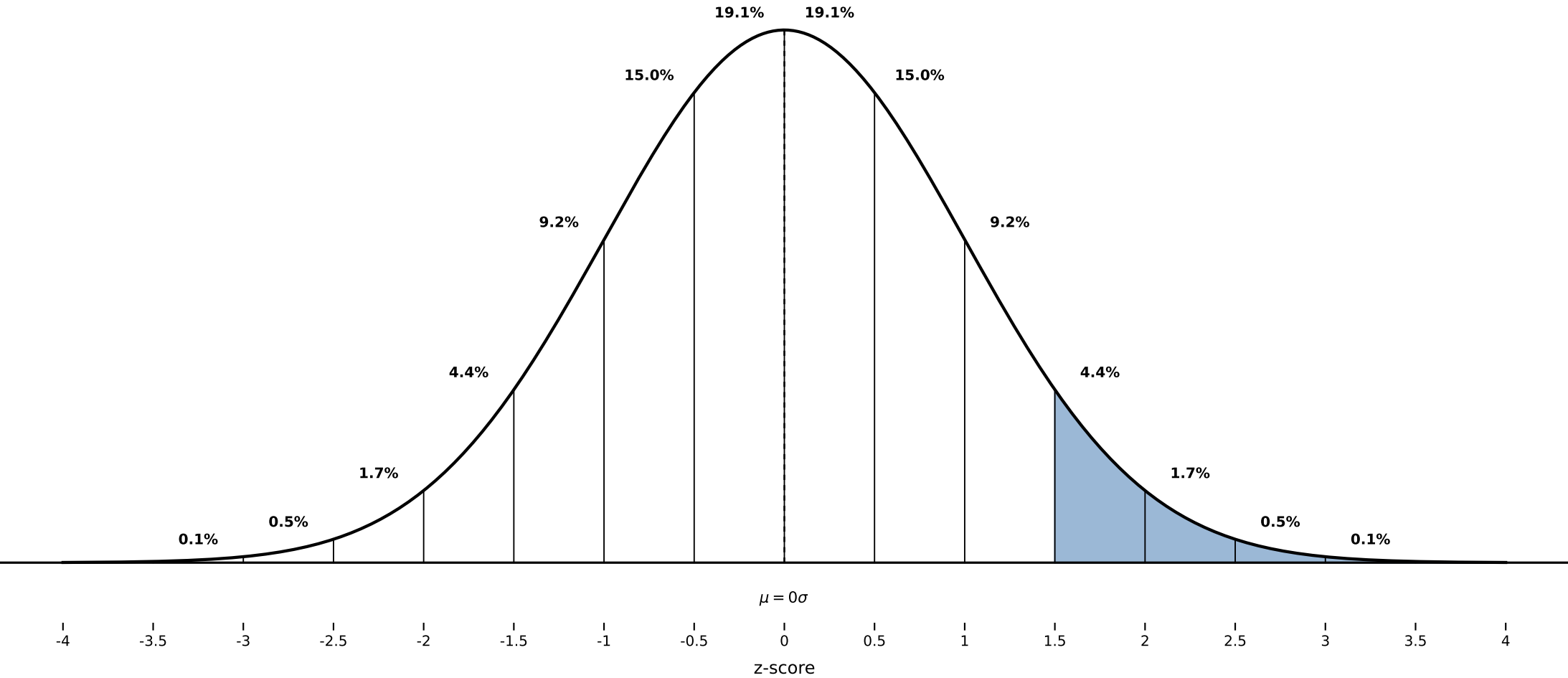
$0.1 + 0.5 + 1.7 + 4.4 + 9.2 = 15.9\%$



(e)(ii) a harvest yield of z-score > 1.5 (X = 4.575 bushels)

$P(Z > 1.5) = 6.7\%$

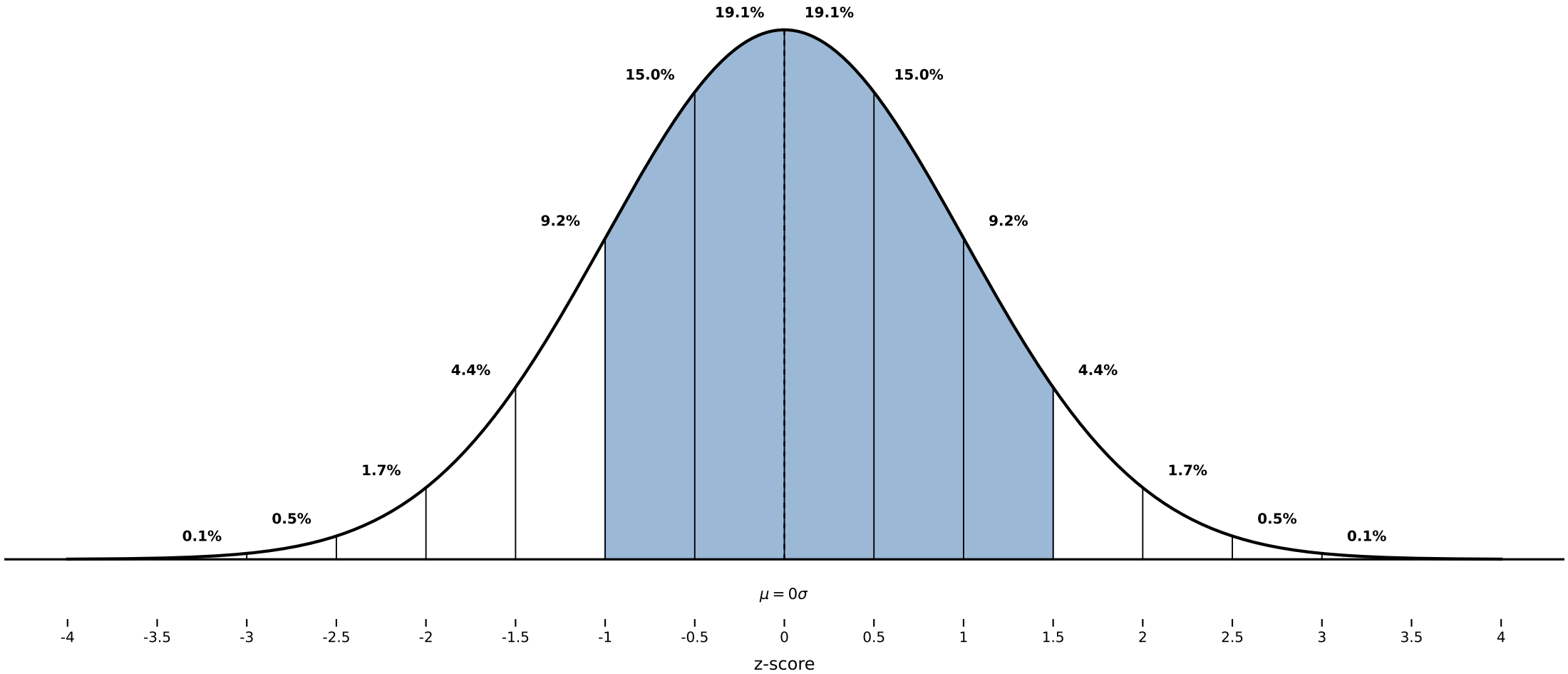
$4.4 + 1.7 + 0.5 + 0.1 = 6.7\%$



(e)(iii) a harvest yield with $-1 < z\text{-score} < 1.5$ (X between 3.45 and 4.575 bushels)

$P(-1 < Z < 1.5) = 77.4\%$

$93.3\% - 15.9\% = 77.4\%$



Question 5(e) — Worked answers

Given

Two crop yields with $z_1 = -1$ and $z_2 = 1.5$, where $\mu = 3.9$ and $\sigma = 0.45$ bushels.

Probabilities

(i) A harvest yield with $z < -1$

Bands from the left tail up to -1 :

$$0.1 + 0.5 + 1.7 + 4.4 + 9.2$$

$$P(Z < -1) = 15.9\%$$

(ii) A harvest yield with $z > 1.5$

Bands from 1.5 out to the right tail:

$$4.4 + 1.7 + 0.5 + 0.1$$

$$P(Z > 1.5) = 6.7\%$$

(iii) A harvest yield with $-1 < z < 1.5$

The bands between the two marks:

$$15.0 + 19.1 + 19.1 + 15.0 + 9.2$$

or $93.3\% - 15.9\%$

$$P(-1 < Z < 1.5) = 77.4\%$$

Check: $15.9\% + 6.7\% + 77.4\% = 100\%$ — the three regions tile the whole distribution.

Actual yields (X-scores)

Rearranging $z = \frac{X - \mu}{\sigma}$ **gives** $X = \mu + z\sigma$

$$z = -1: \quad X = 3.9 + (-1)(0.45) = 3.9 - 0.45$$

$$X = 3.45 \text{ bushels}$$

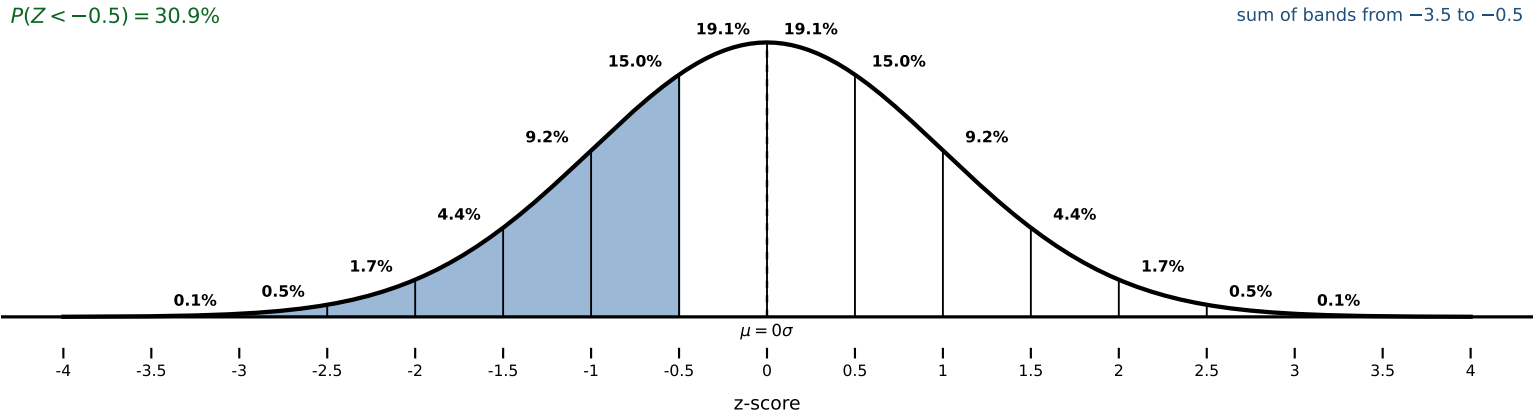
$$z = 1.5: \quad X = 3.9 + (1.5)(0.45) = 3.9 + 0.675$$

$$X = 4.575 \text{ bushels}$$

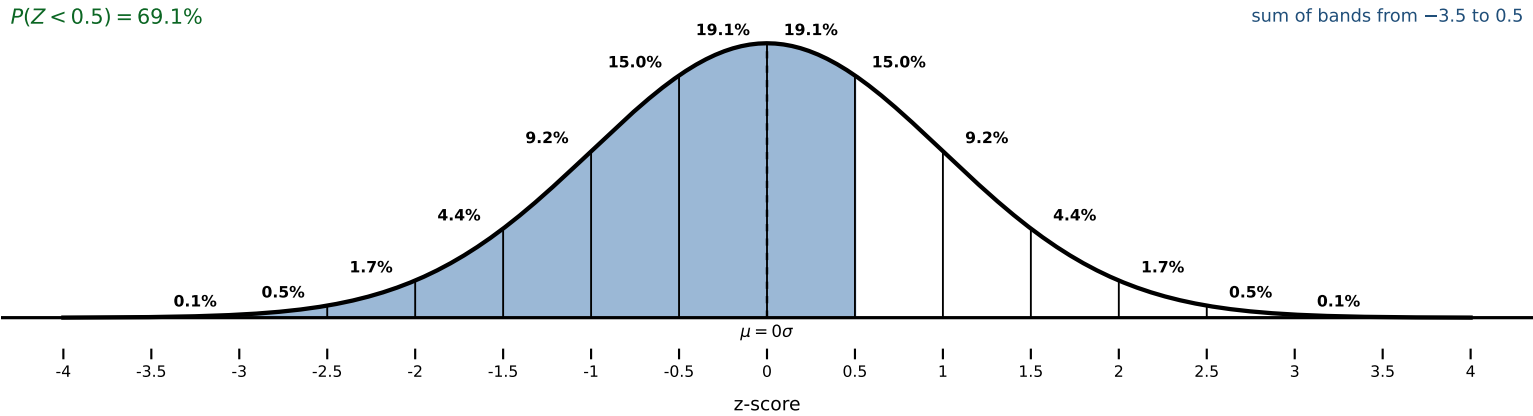
So the two crop yields were 3.45 bushels (one S.D. below μ) and 4.575 bushels (one and a half S.D. above μ).

APPENDIX A — “Left-tailed” for every Q4 z-score
The pairing used in the main answer is $z = -0.5$

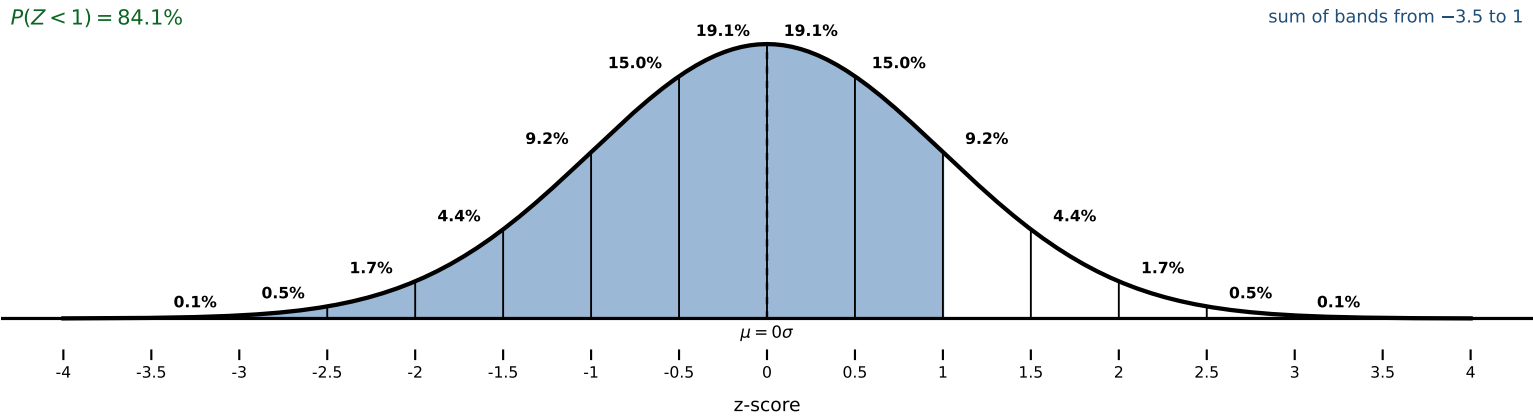
$z = -0.5$ ($X = 3.675$ bushels)



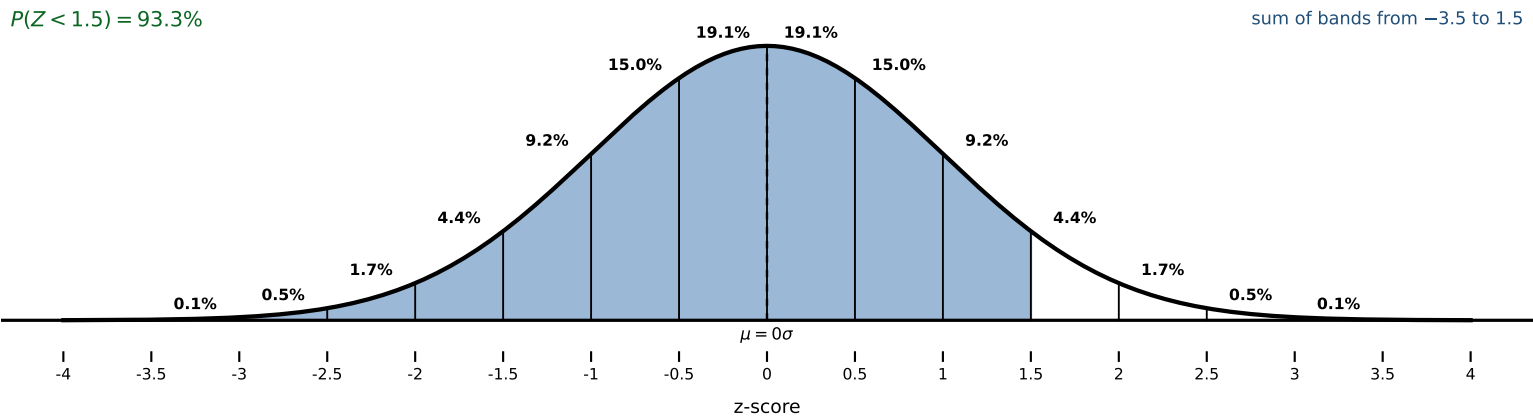
$z = 0.5$ ($X = 4.125$ bushels)



$z = 1$ ($X = 4.35$ bushels)



$z = 1.5$ ($X = 4.575$ bushels)



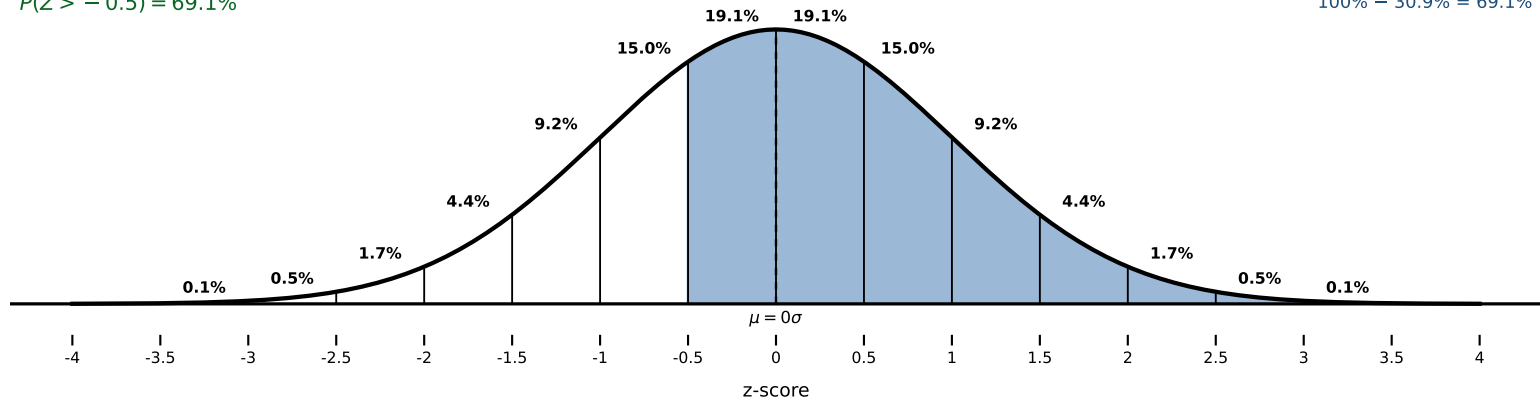
APPENDIX B — “Right-tailed” for every Q4 z-score

The pairing used in the main answer is $z = 1.5$

$z = -0.5$ ($X = 3.675$ bushels)

$$P(Z > -0.5) = 69.1\%$$

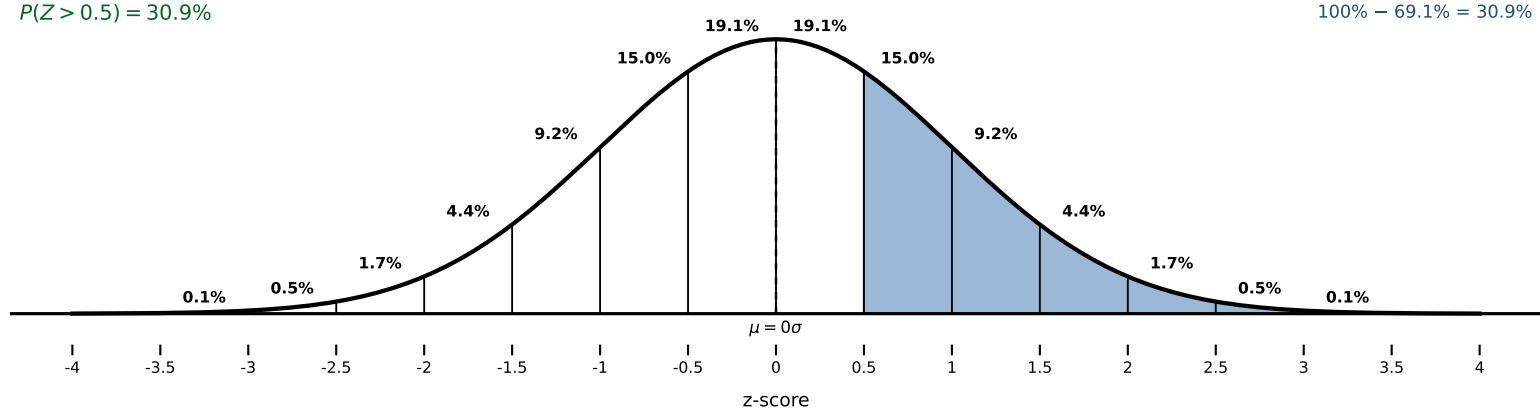
$$100\% - 30.9\% = 69.1\%$$



$z = 0.5$ ($X = 4.125$ bushels)

$$P(Z > 0.5) = 30.9\%$$

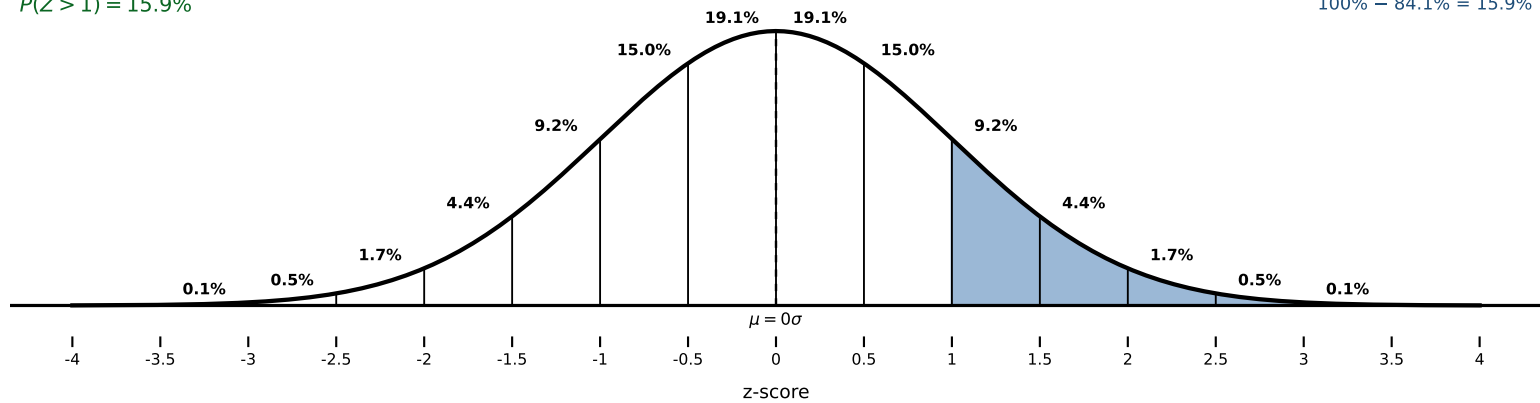
$$100\% - 69.1\% = 30.9\%$$



$z = 1$ ($X = 4.35$ bushels)

$$P(Z > 1) = 15.9\%$$

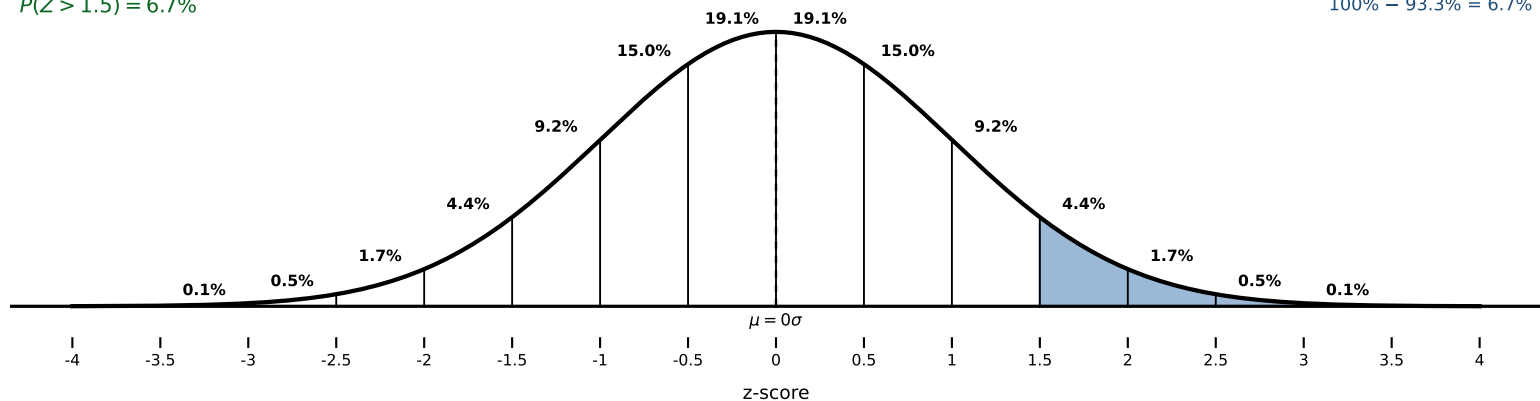
$$100\% - 84.1\% = 15.9\%$$



$z = 1.5$ ($X = 4.575$ bushels)

$$P(Z > 1.5) = 6.7\%$$

$$100\% - 93.3\% = 6.7\%$$

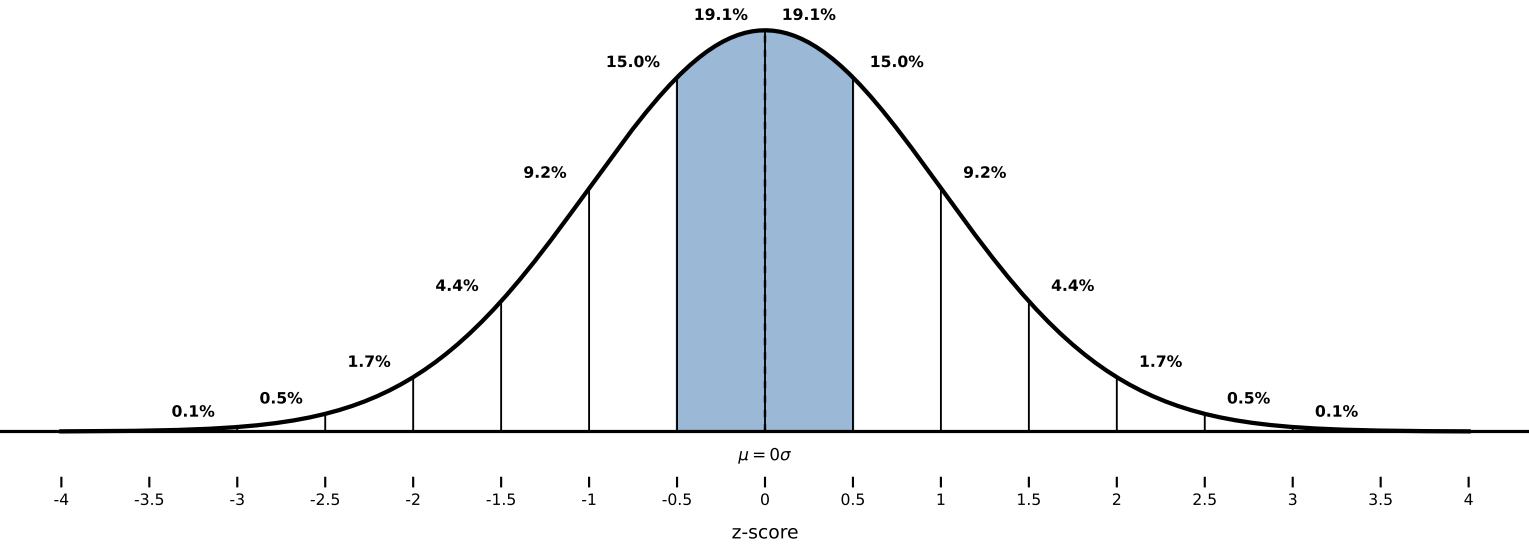


APPENDIX C — “Centred” for every Q4 magnitude
The pairing used in the main answer is $z = \pm 0.5$

$z = \pm 0.5$

$P(-0.5 < Z < 0.5) = 38.2\%$

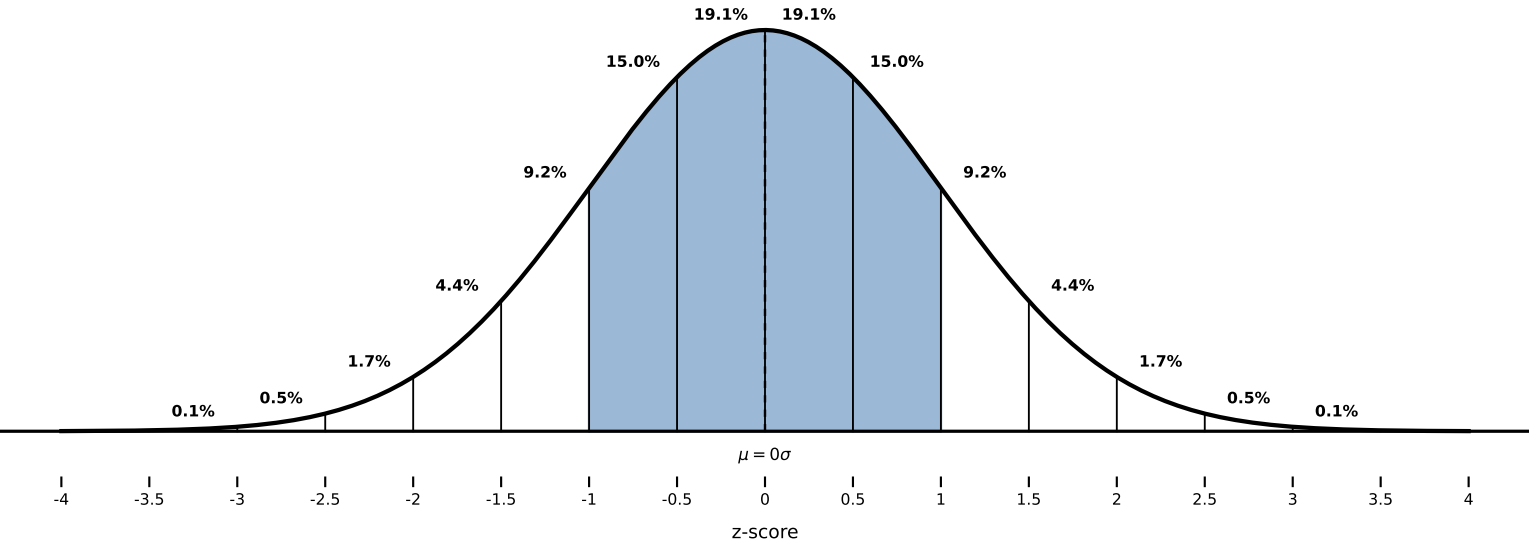
$69.1\% - 30.9\% = 38.2\%$



$z = \pm 1$

$P(-1 < Z < 1) = 68.2\%$

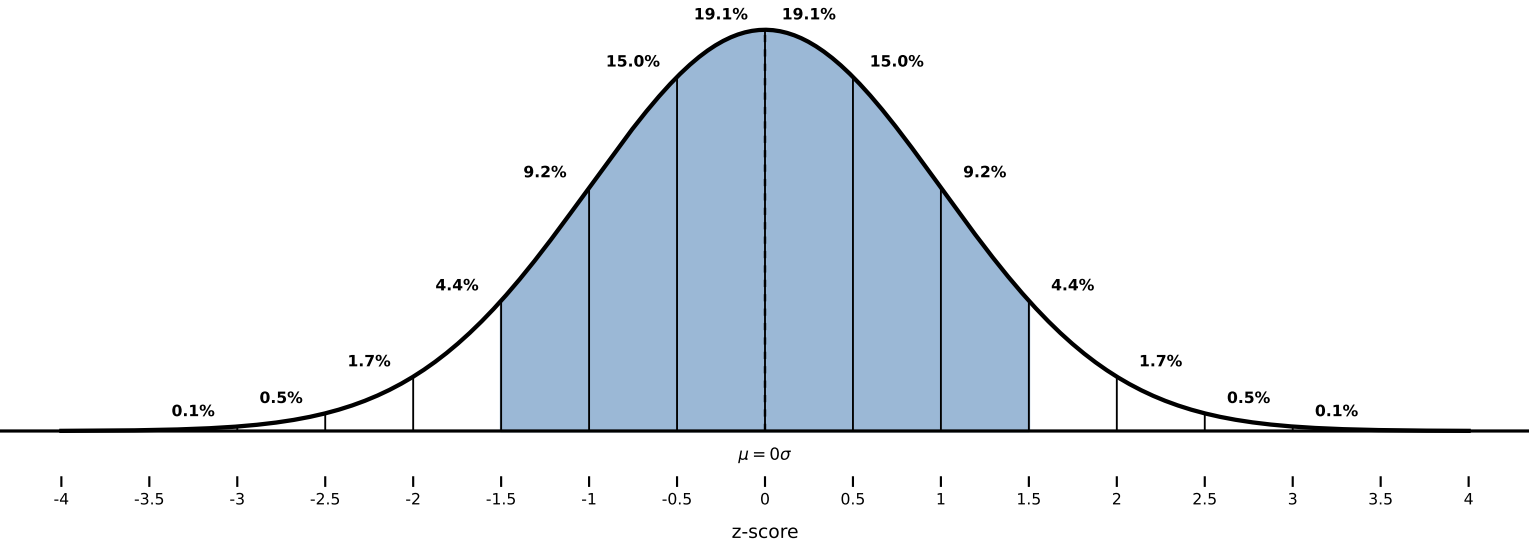
$84.1\% - 15.9\% = 68.2\%$



$z = \pm 1.5$

$P(-1.5 < Z < 1.5) = 86.6\%$

$93.3\% - 6.7\% = 86.6\%$



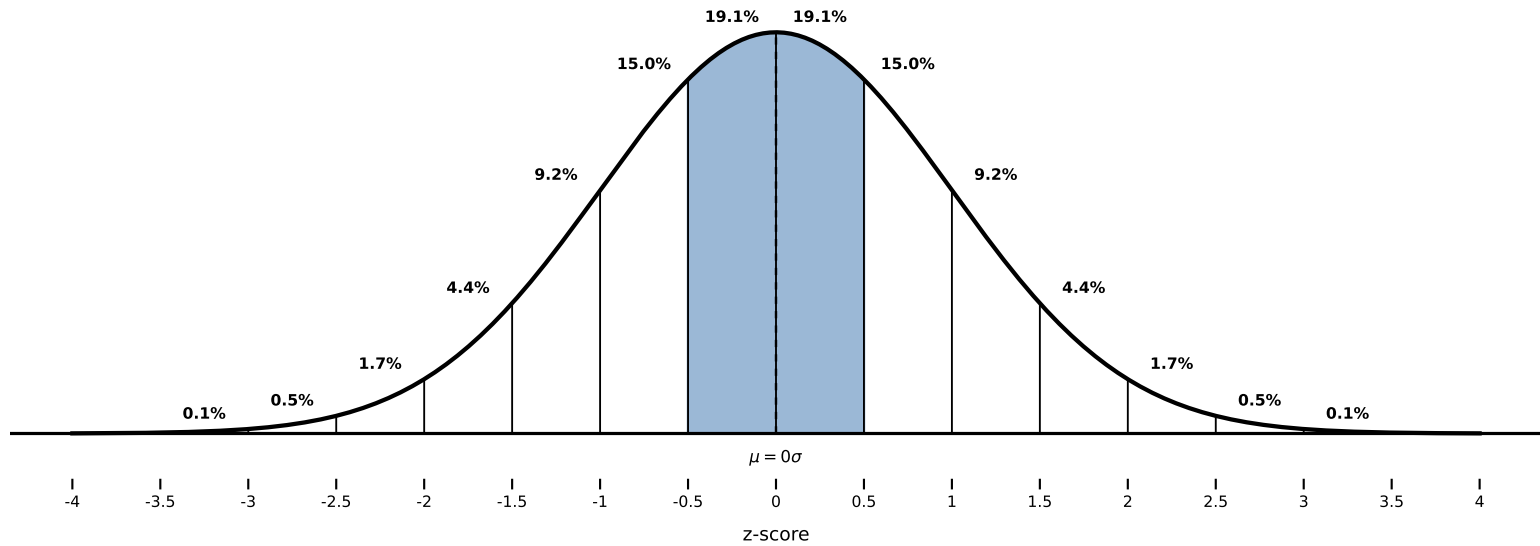
APPENDIX D — “Bounded”, all 6 pairs of Q4 z-scores (1 of 2)

The pairing used in the main answer is $1 < z < 1.5$

$-0.5 < z < 0.5$ (X = 3.675 & 4.125 bushels)

$$P(-0.5 < Z < 0.5) = 38.2\%$$

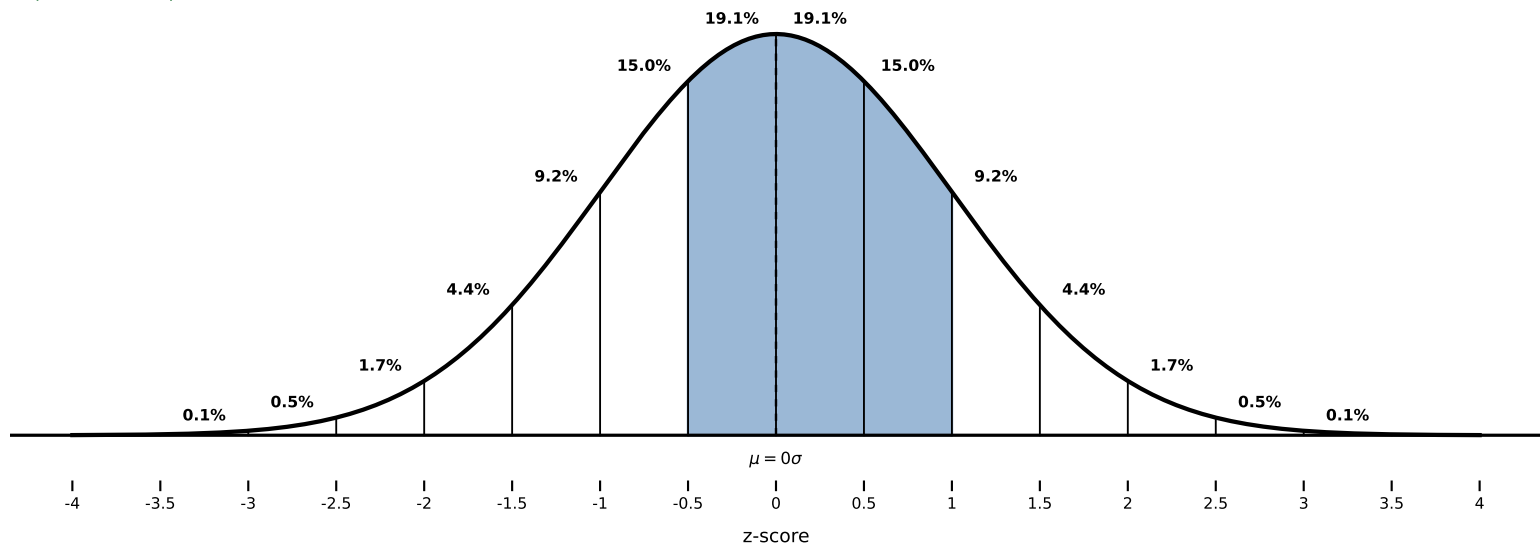
$$69.1\% - 30.9\% = 38.2\%$$



$-0.5 < z < 1$ (X = 3.675 & 4.35 bushels)

$$P(-0.5 < Z < 1) = 53.2\%$$

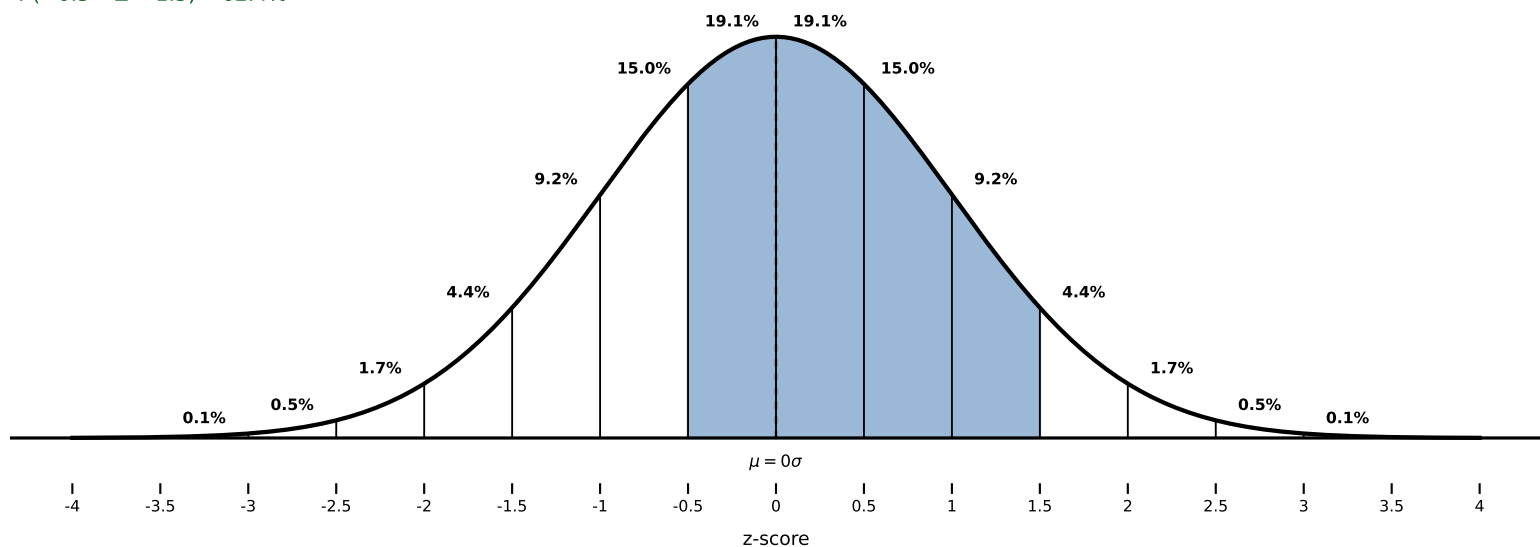
$$84.1\% - 30.9\% = 53.2\%$$



$-0.5 < z < 1.5$ (X = 3.675 & 4.575 bushels)

$$P(-0.5 < Z < 1.5) = 62.4\%$$

$$93.3\% - 30.9\% = 62.4\%$$



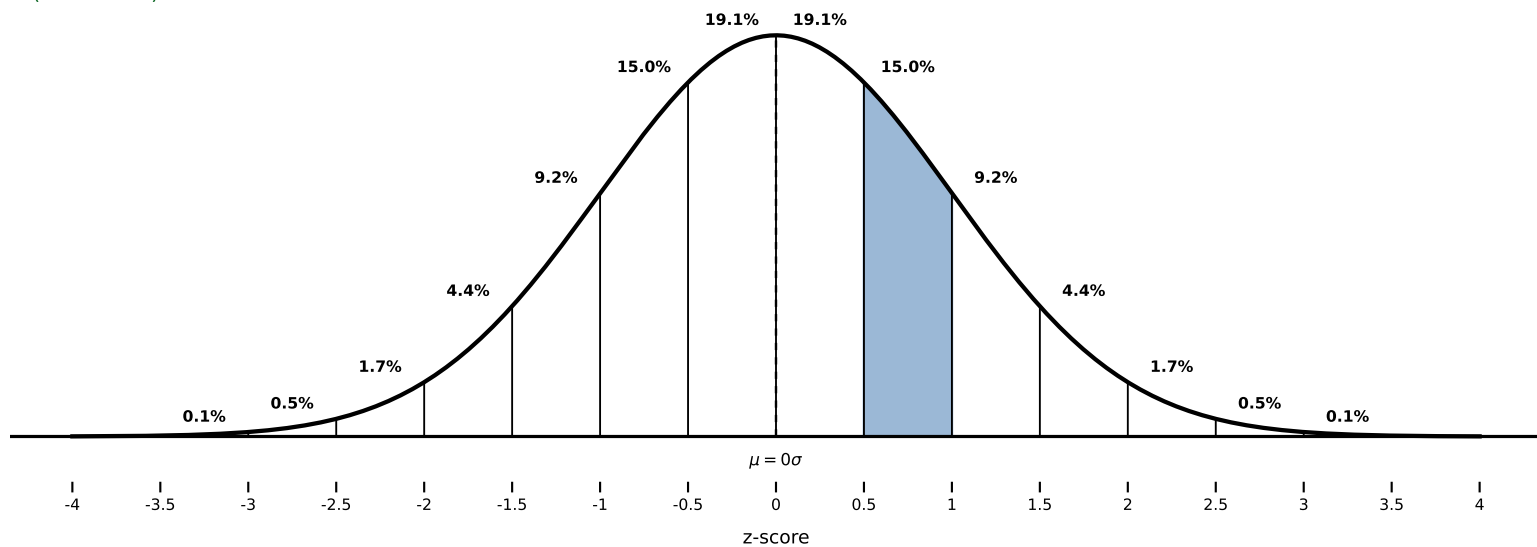
APPENDIX D — “Bounded”, all 6 pairs of Q4 z-scores (2 of 2)

The pairing used in the main answer is $1 < z < 1.5$

$0.5 < z < 1$ (X = 4.125 & 4.35 bushels)

$$P(0.5 < Z < 1) = 15.0\%$$

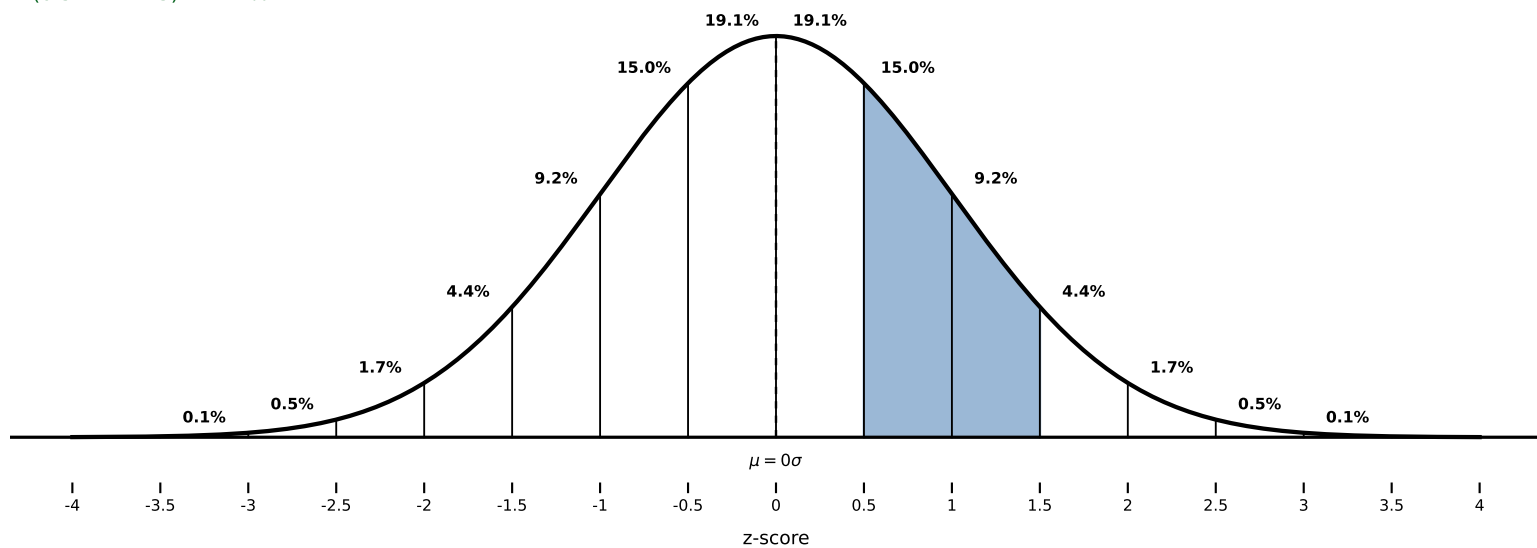
$$84.1\% - 69.1\% = 15.0\%$$



$0.5 < z < 1.5$ (X = 4.125 & 4.575 bushels)

$$P(0.5 < Z < 1.5) = 24.2\%$$

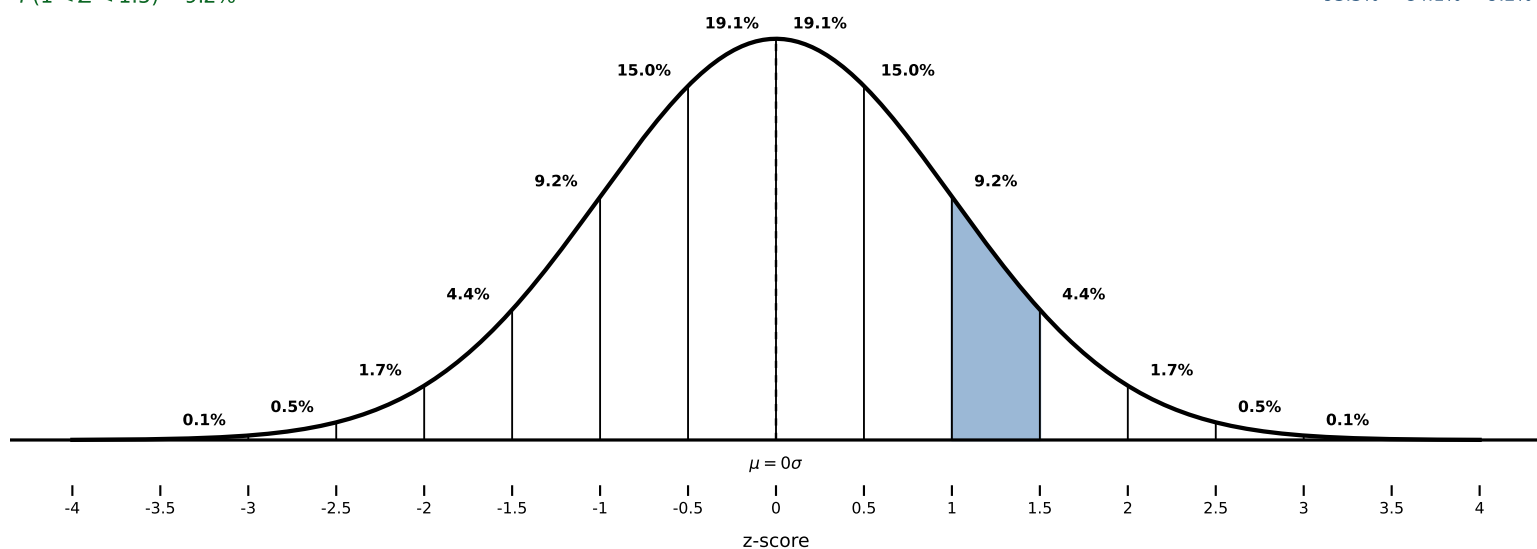
$$93.3\% - 69.1\% = 24.2\%$$



$1 < z < 1.5$ (X = 4.35 & 4.575 bushels)

$$P(1 < Z < 1.5) = 9.2\%$$

$$93.3\% - 84.1\% = 9.2\%$$



APPENDIX E — Master table of every option

Q4 z-scores: -0.5 , $+0.5$, $+1.0$, $+1.5$ (★ = used in the main answer)

Left-tailed $P(Z < z)$

★ $z = -0.5$	30.9%
$z = 0.5$	69.1%
$z = 1$	84.1%
$z = 1.5$	93.3%

Right-tailed $P(Z > z)$

$z = -0.5$	69.1%
$z = 0.5$	30.9%
$z = 1$	15.9%
★ $z = 1.5$	6.7%

Centred $P(-|z| < Z < |z|)$

★ $z = \pm 0.5$	38.2%
$z = \pm 1$	68.2%
$z = \pm 1.5$	86.6%

Bounded $P(z_1 < Z < z_2)$

$-0.5 < z < 0.5$	38.2%
$-0.5 < z < 1$	53.2%
$-0.5 < z < 1.5$	62.4%
$0.5 < z < 1$	15.0%
$0.5 < z < 1.5$	24.2%
★ $1 < z < 1.5$	9.2%

All values read from the $\frac{1}{2}\sigma$ bands printed on the worksheet curve, so they carry $\sim 0.1\%$ rounding.